Quantum bits, or qubits, can exist in superposition states.
Unlike classical bits that are either 0 or 1, a qubit represents a linear combination.
Google Quantum AI demonstrated quantum supremacy using the 53-qubit Sycamore processor in 201

Shor's algorithm can factor large integers exponentially faster than classical algorithms.
Grover's algorithm provides a quadratic speedup for unstructured database search.
Fault-tolerant quantum computing requires surface codes with thousands of physical qubits per logical